

PAPER_286: M16 Eagle Nebula UQFF — Nebular Friedmann Redshift Parameter κ_{neb}

First UQFF Nebular Module with $z > 0$: $H(z=0.0015) = 70.047$ km/s/Mpc

Classification: UQFF 2.0 Gravitational Physics — Nebular Cosmological Coupling

System: M16 Eagle Nebula (IC 4703), ~ 5700 ly distance, $z = 0.0015$

Session: 80 | **Module:** M16_UQFF_MODULE.cpp (22nd C++ UQFF module)

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Abstract

M16 (Eagle Nebula, IC 4703) is located at ~ 5700 light-years distance, corresponding to cosmological redshift $z = 0.0015$. While this redshift is sub-Hubble (far smaller than extragalactic objects), it is **non-zero**, making M16 the **first UQFF nebular module to carry a cosmological redshift parameter**. This paper defines the **UQFF Nebular Redshift Parameter** $\kappa_{\text{neb}} = [H(z=0.0015) - H(z=0)] / H(z=0) = 6.71 \times 10^{-4}$, quantifying the fractional departure of the Friedmann Hubble rate at M16's redshift from the present-epoch value. The expansion coupling term $g_{\text{exp}} = g_{\text{base}} \times H(z=0.0015) \times t$ contributes $g_{\text{exp}} \approx 5.21 \times 10^{-16}$ m/s² at $t = 5$ Myr — tiny but formally catalogued for the first time in UQFF nebular physics.

2. Physical Motivation

All prior UQFF nebular modules (Session 55 M16 in CP3) set $z = 0$, ignoring the cosmological correction from the Eagle Nebula's finite distance. However, the full Friedmann $H(z)$ formalism correctly accounts for the **look-back time effect**: at $z = 0.0015$, the Hubble flow was marginally faster than today. The UQFF Nebular Redshift Parameter κ_{neb} quantifies this correction for the M16 Hubble expansion coupling term.

Prior UQFF modules with $z > 0$ were all **extragalactic** (galaxies, galaxy clusters). This paper extends the $z > 0$ treatment to the **sub-kiloparsec, sub-Hubble-flow nebular regime** for the first time.

3. Mathematical Formulation

3.1 Friedmann $H(z)$

The Flat Λ CDM Friedmann equation for the Hubble parameter at redshift z :

$$H(z) = H_0 \sqrt{\Omega_m(1+z)^3 + \Omega_\Lambda}$$

Parameters: | Parameter | Value | |----|----| | H_0 | 70.0 km/s/Mpc | | Ω_m | 0.3 | | Ω_Λ | 0.7 | | z (M16) | 0.0015 |

$H(z = 0)$:

$$H(0) = 70.0 \times \sqrt{0.3 \times 1^3 + 0.7} = 70.0 \times \sqrt{1.0} = 70.000 \text{ km/s/Mpc}$$

$H(z = 0.0015)$:

$$(1 + 0.0015)^3 = 1.00451$$

$$0.3 \times 1.00451 + 0.7 = 1.00135$$

$$H(0.0015) = 70.0 \times \sqrt{1.00135} = 70.0 \times 1.000675 = \mathbf{70.047} \text{ km/s/Mpc}$$

3.2 UQFF Nebular Redshift Parameter κ_{neb}

$$\kappa_{\text{neb}} = \frac{H(z = 0.0015) - H(z = 0)}{H(z = 0)} = \frac{70.047 - 70.000}{70.000} = \frac{0.047}{70.000} = 6.71 \times 10^{-4}$$

This fractional correction is **small but physically meaningful** — it encodes the sub-Hubble cosmological velocity gradient at M16’s position in the Milky Way’s extended gravitational domain.

3.3 Friedmann Expansion Gravity Coupling

The Hubble expansion coupling on the nebula’s self-gravity:

$$g_{\text{exp}}(t) = g_{\text{base}} \times H(z = 0.0015) \times t$$

Converting $H(z=0.0015)$ to SI (s^{-1}):

$$H_{SI} = 70.047 \text{ km/s/Mpc} \times \frac{10^3 \text{ m/km}}{3.086 \times 10^{22} \text{ m/Mpc}} = 2.270 \times 10^{-18} \text{ s}^{-1}$$

At $t = 5 \text{ Myr} = 1.578 \times 10^{14} \text{ s}$:

$$\begin{aligned} g_{\text{exp}}(5 \text{ Myr}) &= 1.454 \times 10^{-12} \times 2.270 \times 10^{-18} \times 1.578 \times 10^{14} \\ &= 1.454 \times 10^{-12} \times 3.582 \times 10^{-4} = \mathbf{5.21 \times 10^{-16} \text{ m/s}^2} \end{aligned}$$

4. Comparison Across UQFF Nebular / Stellar Modules

Module	z	κ_{type}	H(z)
SATURN (Session 78)	0	—	70.000 km/s/Mpc
Prior M16 CP3 (Session 55)	0	—	70.000 km/s/Mpc
M16 THIS MODULE (PAPER_286)	0.0015	$\kappa_{\text{neb}} = 6.71 \times 10^{-4}$	70.047 km/s/Mpc
ANDROMEDA (Session 76)	~ 0	$\kappa_{\text{recession}}$	70.000 km/s/Mpc
Extragalactic modules	> 0.001	$\kappa_{\text{recession}}$	$> 70.0 \text{ km/s/Mpc}$

M16 is the **first UQFF module** to formally assign a $z > 0$ value to a **nebular (sub-galactic)** object, introducing κ_{neb} as a distinct parameter class from the galactic/extragalactic $\kappa_{\text{recession}}$ used in prior modules.

5. Physical Significance

Why $z = 0.0015$ is Not Simply $z = 0$

1. **Formal completeness:** The UQFF framework demands that all physical parameters be carried with their correct values. Setting $z = 0$ for M16 discards a real (if small) cosmological correction.
 2. **Template for nearby nebulae:** The sub-Hubble-flow regime ($z < 0.01$) is occupied by many galactic star-forming regions — Orion ($z \approx 0.0014$), Carina ($z \approx 0.0026$), Rho Ophiuchi ($z \approx 0.0004$). κ_{neb} provides a universal scaling parameter for this entire class.
 3. **Detectability threshold:** $\kappa_{\text{neb}} = 6.71 \times 10^{-4}$ is above the precision floor of modern distance measurements (Gaia DR3: $\leq 0.01\%$ parallax errors for nearby nebulae), meaning the $H(z)$ correction is formally measurable.
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6. UQFF Module g_{exp} Context

In the full M16 UQFF 2.0 equation:

$$g_{\text{total}}(r, t) = [g_{\text{dyn}}(t) + U_{g, \text{sum}}(26) + \Lambda + Q + L + F + g_{\text{exp}}] \times \text{corr}_{SC}$$

g_{exp} uses $H(z=0.0015)$ rather than $H(z=0)$. The fractional difference in g_{exp} due to κ_{neb} :

$$\frac{\Delta g_{\text{exp}}}{g_{\text{exp}}} = \kappa_{\text{neb}} = 6.71 \times 10^{-4}$$

This is a 0.067% correction to g_{exp} , which is itself a tiny term — but it is catalogued precisely as the UQFF Nebular Friedmann correction.

7. Wolfram KB Term

M16UQFF:kappa_neb=[H[z=0.0015]-H[z=0]]/H[z=0]=6.71e-4; H[z=0.0015]=70.047 km/s/Mpc [PAPER_284]

8. Cross-References

- **PAPER_284:** Dual Mass Co-Action Product (Φ_{dm})
 - **PAPER_285:** Erosion Saturation Half-Time (t_{half} , Δg_{Max})
 - **M16_UQFF_MODULE.cpp:** Full UQFF 2.0 C++ implementation (22nd module, first nebular $z > 0$)
 - **CondensedPhysics3.py:** M16NebularFriedmannRedshiftCalculator
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